

HỌC VIỆN CÔNG NGHỆ BƯU CHÍNH VIỄN THÔNG
KHOA CÔNG NGHỆ THÔNG TIN



BÁO CÁO ĐỀ CƯƠNG THỰC TẬP CƠ SỞ

ĐỀ TÀI:

**Nghiên cứu học đa nhiệm và cơ chế Attention trong bài toán
phân đoạn và phân loại u não từ ảnh MRI**

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HÀ NỘI, THÁNG 4/2026

Báo cáo tuần 6

I) Công việc hoàn thành

Thực hiện train 3 model UnetClassifier, Unet, và Multitask Learning với 40 epochs

1. Train Model

1.1 Classification

Running on cuda

HYPERPARAMETERS SUMMARY

LR_CLF : 0.0001
LR_SEG : 0.001
BATCH_CLF : 32
BATCH_SEG : 16
BATCH_JOINT : 12
EPOCHS : 40
OPTIMIZER : AdamW
LOSS_CLF : CrossEntropyLoss
LOSS_SEG : BCEWithLogitsLoss (Weighted) + DiceLoss
EARLY_STOP_PATIENCE : 10
WEIGHT_DECAY : 0.001

CLASSIFICATION TRAINING

Epoch 1: Train Acc=0.4300 | Val Acc=0.4050
Saved best model (Val Acc: 0.4050)
Epoch 2: Train Acc=0.4928 | Val Acc=0.4900
Saved best model (Val Acc: 0.4900)
Epoch 3: Train Acc=0.5597 | Val Acc=0.6050
Saved best model (Val Acc: 0.6050)
Epoch 4: Train Acc=0.6188 | Val Acc=0.4110
Epoch 5: Train Acc=0.6633 | Val Acc=0.6080
Saved best model (Val Acc: 0.6080)
Epoch 6: Train Acc=0.7077 | Val Acc=0.6680
Saved best model (Val Acc: 0.6680)
Epoch 7: Train Acc=0.7368 | Val Acc=0.6570
Epoch 8: Train Acc=0.7515 | Val Acc=0.4490
Epoch 9: Train Acc=0.7598 | Val Acc=0.6100
Epoch 10: Train Acc=0.7718 | Val Acc=0.6310
Epoch 11: Train Acc=0.7930 | Val Acc=0.7790
Saved best model (Val Acc: 0.7790)
Epoch 12: Train Acc=0.7935 | Val Acc=0.6090
Epoch 13: Train Acc=0.8007 | Val Acc=0.7870
Saved best model (Val Acc: 0.7870)
Epoch 14: Train Acc=0.8103 | Val Acc=0.7200

Epoch 15: Train Acc=0.8265 | Val Acc=0.7330
Epoch 16: Train Acc=0.8245 | Val Acc=0.8200
Saved best model (Val Acc: 0.8200)
Epoch 17: Train Acc=0.8245 | Val Acc=0.7920
Epoch 18: Train Acc=0.8365 | Val Acc=0.8290
Saved best model (Val Acc: 0.8290)
Epoch 19: Train Acc=0.8458 | Val Acc=0.8410
Saved best model (Val Acc: 0.8410)
Epoch 20: Train Acc=0.8538 | Val Acc=0.7770
Epoch 21: Train Acc=0.8522 | Val Acc=0.7240
Epoch 22: Train Acc=0.8612 | Val Acc=0.7700
Epoch 23: Train Acc=0.8645 | Val Acc=0.8410
Epoch 24: Train Acc=0.8710 | Val Acc=0.8670
Saved best model (Val Acc: 0.8670)
Epoch 25: Train Acc=0.8698 | Val Acc=0.8380
Epoch 26: Train Acc=0.8702 | Val Acc=0.8100
Epoch 27: Train Acc=0.8815 | Val Acc=0.7650
Epoch 28: Train Acc=0.8775 | Val Acc=0.8450
Epoch 29: Train Acc=0.8758 | Val Acc=0.7450
Epoch 30: Train Acc=0.8802 | Val Acc=0.8690
Saved best model (Val Acc: 0.8690)
Epoch 31: Train Acc=0.8765 | Val Acc=0.7800
Epoch 32: Train Acc=0.8900 | Val Acc=0.8600
Epoch 33: Train Acc=0.8912 | Val Acc=0.8590
Epoch 34: Train Acc=0.8840 | Val Acc=0.8650
Epoch 35: Train Acc=0.8860 | Val Acc=0.8330
Epoch 36: Train Acc=0.8978 | Val Acc=0.8970
Saved best model (Val Acc: 0.8970)
Epoch 37: Train Acc=0.8880 | Val Acc=0.8230
Epoch 38: Train Acc=0.8890 | Val Acc=0.8530
Epoch 39: Train Acc=0.8878 | Val Acc=0.8220
Epoch 40: Train Acc=0.8932 | Val Acc=0.8430

EVALUATING CLASSIFICATION ON TEST SET...

Test Acc: 0.9020 | Prec: 0.9067 | Recall: 0.9020 | F1: 0.9004

1.2 Segmentation using Base Unet

Running on cuda

HYPERPARAMETERS SUMMARY

LR_CLF : 0.0001

LR_SEG : 0.001
BATCH_CLF : 32
BATCH_SEG : 16
BATCH_JOINT : 12
EPOCHS : 40
OPTIMIZER : AdamW
LOSS_CLF : CrossEntropyLoss
LOSS_SEG : BCEWithLogitsLoss (Weighted) + DiceLoss
EARLY_STOP_PATIENCE : 10
WEIGHT_DECAY : 0.001
SEGMENTATION TRAINING
Epoch 1: Val mIoU=0.0697 | Dice=0.1285 | Pixel Acc=0.9598
Saved best model (Val mIoU: 0.0697)
Epoch 2: Val mIoU=0.1391 | Dice=0.2422 | Pixel Acc=0.9525
Saved best model (Val mIoU: 0.1391)
Epoch 3: Val mIoU=0.0378 | Dice=0.0724 | Pixel Acc=0.8773
Epoch 4: Val mIoU=0.2527 | Dice=0.3968 | Pixel Acc=0.9829
Saved best model (Val mIoU: 0.2527)
Epoch 5: Val mIoU=0.2469 | Dice=0.3916 | Pixel Acc=0.9770
Epoch 6: Val mIoU=0.2300 | Dice=0.3693 | Pixel Acc=0.9662
Epoch 7: Val mIoU=0.3356 | Dice=0.4976 | Pixel Acc=0.9859
Saved best model (Val mIoU: 0.3356)
Epoch 8: Val mIoU=0.4023 | Dice=0.5677 | Pixel Acc=0.9874
Saved best model (Val mIoU: 0.4023)
Epoch 9: Val mIoU=0.3952 | Dice=0.5609 | Pixel Acc=0.9846
Epoch 10: Val mIoU=0.4308 | Dice=0.5981 | Pixel Acc=0.9877
Saved best model (Val mIoU: 0.4308)
Epoch 11: Val mIoU=0.4662 | Dice=0.6293 | Pixel Acc=0.9881
Saved best model (Val mIoU: 0.4662)
Epoch 12: Val mIoU=0.5107 | Dice=0.6692 | Pixel Acc=0.9906
Saved best model (Val mIoU: 0.5107)
Epoch 13: Val mIoU=0.5180 | Dice=0.6760 | Pixel Acc=0.9902
Saved best model (Val mIoU: 0.5180)
Epoch 14: Val mIoU=0.4794 | Dice=0.6415 | Pixel Acc=0.9905
Epoch 15: Val mIoU=0.5237 | Dice=0.6817 | Pixel Acc=0.9909
Saved best model (Val mIoU: 0.5237)
Epoch 16: Val mIoU=0.4823 | Dice=0.6433 | Pixel Acc=0.9910
Epoch 17: Val mIoU=0.4797 | Dice=0.6444 | Pixel Acc=0.9854
Epoch 18: Val mIoU=0.5579 | Dice=0.7124 | Pixel Acc=0.9903
Saved best model (Val mIoU: 0.5579)

Epoch 19: Val mIoU=0.5368 | Dice=0.6917 | Pixel Acc=0.9917
Epoch 20: Val mIoU=0.5625 | Dice=0.7164 | Pixel Acc=0.9902
Saved best model (Val mIoU: 0.5625)
Epoch 21: Val mIoU=0.5758 | Dice=0.7257 | Pixel Acc=0.9913
Saved best model (Val mIoU: 0.5758)
Epoch 22: Val mIoU=0.5627 | Dice=0.7146 | Pixel Acc=0.9913
Epoch 23: Val mIoU=0.5669 | Dice=0.7180 | Pixel Acc=0.9908
Epoch 24: Val mIoU=0.5927 | Dice=0.7394 | Pixel Acc=0.9917
Saved best model (Val mIoU: 0.5927)
Epoch 25: Val mIoU=0.5848 | Dice=0.7333 | Pixel Acc=0.9914
Epoch 26: Val mIoU=0.5689 | Dice=0.7196 | Pixel Acc=0.9897
Epoch 27: Val mIoU=0.6141 | Dice=0.7567 | Pixel Acc=0.9925
Saved best model (Val mIoU: 0.6141)
Epoch 28: Val mIoU=0.6055 | Dice=0.7485 | Pixel Acc=0.9922
Epoch 29: Val mIoU=0.5622 | Dice=0.7134 | Pixel Acc=0.9922
Epoch 30: Val mIoU=0.6204 | Dice=0.7613 | Pixel Acc=0.9924
Saved best model (Val mIoU: 0.6204)
Epoch 31: Val mIoU=0.6189 | Dice=0.7607 | Pixel Acc=0.9926
Epoch 32: Val mIoU=0.6218 | Dice=0.7622 | Pixel Acc=0.9926
Saved best model (Val mIoU: 0.6218)
Epoch 33: Val mIoU=0.6311 | Dice=0.7694 | Pixel Acc=0.9929
Saved best model (Val mIoU: 0.6311)
Epoch 34: Val mIoU=0.6149 | Dice=0.7577 | Pixel Acc=0.9923
Epoch 35: Val mIoU=0.6275 | Dice=0.7671 | Pixel Acc=0.9931
Epoch 36: Val mIoU=0.5071 | Dice=0.6669 | Pixel Acc=0.9882
Epoch 37: Val mIoU=0.6394 | Dice=0.7766 | Pixel Acc=0.9934
Saved best model (Val mIoU: 0.6394)
Epoch 38: Val mIoU=0.6329 | Dice=0.7716 | Pixel Acc=0.9928
Epoch 39: Val mIoU=0.6425 | Dice=0.7798 | Pixel Acc=0.9930
Saved best model (Val mIoU: 0.6425)
Epoch 40: Val mIoU=0.6430 | Dice=0.7794 | Pixel Acc=0.9926
Saved best model (Val mIoU: 0.6430)

EVALUATING UNET-SEGMENTATION ON TEST SET...

Test mIoU: 0.7130 | Dice: 0.8248 | Pixel Acc: 0.9939

1.3 Multitask Learning

Running on cuda

HYPERPARAMETERS SUMMARY

LR_CLF : 0.0001

LR_SEG : 0.001
BATCH_CLF : 32
BATCH_SEG : 16
BATCH_JOINT : 12
EPOCHS : 40
OPTIMIZER : AdamW
LOSS_CLF : CrossEntropyLoss
LOSS_SEG : BCEWithLogitsLoss (Weighted) + DiceLoss
EARLY_STOP_PATIENCE : 10
WEIGHT_DECAY : 0.001

JOINT TRAINING

Epoch 1: Clf Acc=0.4084 | mIoU=0.1383 | Dice=0.2381 | PixAcc=0.9708
Saved best model (Combined Score: 0.2733)
Epoch 2: Clf Acc=0.5314 | mIoU=0.1650 | Dice=0.2770 | PixAcc=0.9751
Saved best model (Combined Score: 0.3482)
Epoch 3: Clf Acc=0.5655 | mIoU=0.2311 | Dice=0.3667 | PixAcc=0.9857
Saved best model (Combined Score: 0.3983)
Epoch 4: Clf Acc=0.5330 | mIoU=0.2172 | Dice=0.3540 | PixAcc=0.9613
Epoch 5: Clf Acc=0.6181 | mIoU=0.3571 | Dice=0.5188 | PixAcc=0.9870
Saved best model (Combined Score: 0.4876)
Epoch 6: Clf Acc=0.5958 | mIoU=0.3413 | Dice=0.5039 | PixAcc=0.9784
Epoch 7: Clf Acc=0.6176 | mIoU=0.1648 | Dice=0.2738 | PixAcc=0.9862
Epoch 8: Clf Acc=0.6986 | mIoU=0.4742 | Dice=0.6375 | PixAcc=0.9884
Saved best model (Combined Score: 0.5864)
Epoch 9: Clf Acc=0.6699 | mIoU=0.4749 | Dice=0.6366 | PixAcc=0.9899
Epoch 10: Clf Acc=0.7394 | mIoU=0.4496 | Dice=0.6151 | PixAcc=0.9855
Saved best model (Combined Score: 0.5945)
Epoch 11: Clf Acc=0.7713 | mIoU=0.4870 | Dice=0.6487 | PixAcc=0.9880
Saved best model (Combined Score: 0.6291)
Epoch 12: Clf Acc=0.7478 | mIoU=0.5131 | Dice=0.6733 | PixAcc=0.9900
Saved best model (Combined Score: 0.6305)
Epoch 13: Clf Acc=0.7664 | mIoU=0.4589 | Dice=0.6207 | PixAcc=0.9906
Epoch 14: Clf Acc=0.8131 | mIoU=0.5627 | Dice=0.7147 | PixAcc=0.9909
Saved best model (Combined Score: 0.6879)
Epoch 15: Clf Acc=0.8299 | mIoU=0.5557 | Dice=0.7085 | PixAcc=0.9908
Saved best model (Combined Score: 0.6928)
Epoch 16: Clf Acc=0.7523 | mIoU=0.5745 | Dice=0.7243 | PixAcc=0.9917
Epoch 17: Clf Acc=0.8826 | mIoU=0.5778 | Dice=0.7280 | PixAcc=0.9908
Saved best model (Combined Score: 0.7302)
Epoch 18: Clf Acc=0.8611 | mIoU=0.5859 | Dice=0.7340 | PixAcc=0.9910

Epoch 19: Clf Acc=0.8699 | mIoU=0.5925 | Dice=0.7401 | PixAcc=0.9916
Saved best model (Combined Score: 0.7312)
Epoch 20: Clf Acc=0.8476 | mIoU=0.6019 | Dice=0.7463 | PixAcc=0.9919
Epoch 21: Clf Acc=0.8021 | mIoU=0.6111 | Dice=0.7549 | PixAcc=0.9923
Epoch 22: Clf Acc=0.8770 | mIoU=0.6311 | Dice=0.7707 | PixAcc=0.9928
Saved best model (Combined Score: 0.7540)
Epoch 23: Clf Acc=0.8438 | mIoU=0.6135 | Dice=0.7563 | PixAcc=0.9922
Epoch 24: Clf Acc=0.8838 | mIoU=0.6185 | Dice=0.7597 | PixAcc=0.9930
Epoch 25: Clf Acc=0.8644 | mIoU=0.6181 | Dice=0.7607 | PixAcc=0.9919
Epoch 26: Clf Acc=0.8775 | mIoU=0.6314 | Dice=0.7704 | PixAcc=0.9932
Saved best model (Combined Score: 0.7545)
Epoch 27: Clf Acc=0.8851 | mIoU=0.6154 | Dice=0.7574 | PixAcc=0.9925
Epoch 28: Clf Acc=0.8889 | mIoU=0.6298 | Dice=0.7694 | PixAcc=0.9924
Saved best model (Combined Score: 0.7593)
Epoch 29: Clf Acc=0.8824 | mIoU=0.6213 | Dice=0.7628 | PixAcc=0.9924
Epoch 30: Clf Acc=0.8902 | mIoU=0.5850 | Dice=0.7346 | PixAcc=0.9899
Epoch 31: Clf Acc=0.9192 | mIoU=0.6417 | Dice=0.7774 | PixAcc=0.9926
Saved best model (Combined Score: 0.7804)
Epoch 32: Clf Acc=0.9003 | mIoU=0.6348 | Dice=0.7727 | PixAcc=0.9928
Epoch 33: Clf Acc=0.9095 | mIoU=0.6135 | Dice=0.7560 | PixAcc=0.9913
Epoch 34: Clf Acc=0.8716 | mIoU=0.6489 | Dice=0.7834 | PixAcc=0.9934
Epoch 35: Clf Acc=0.7605 | mIoU=0.6094 | Dice=0.7544 | PixAcc=0.9911
Epoch 36: Clf Acc=0.9192 | mIoU=0.6400 | Dice=0.7752 | PixAcc=0.9934
Epoch 37: Clf Acc=0.8990 | mIoU=0.6602 | Dice=0.7915 | PixAcc=0.9934
Epoch 38: Clf Acc=0.9179 | mIoU=0.6723 | Dice=0.8000 | PixAcc=0.9936
Saved best model (Combined Score: 0.7951)
Epoch 39: Clf Acc=0.8741 | mIoU=0.6554 | Dice=0.7878 | PixAcc=0.9931
Epoch 40: Clf Acc=0.9306 | mIoU=0.6690 | Dice=0.7983 | PixAcc=0.9935
Saved best model (Combined Score: 0.7998)

Generating Joint Model Confusion Matrix...

EVALUATING JOINT MODEL ON TEST SET...

Test Clf Acc: 0.9384 | mIoU: 0.7382 | Dice: 0.8415 | Pixel Acc: 0.9948

2. Dựng giao diện để demo

- Làm giao diện upload ảnh -> Sinh hình ảnh dự đoán kết quả của 4 model để so sánh

II) Nhiệm vụ tuần sau

1. Train Attention Unet với 40 epochs
2. Hoàn thiện nốt dự án